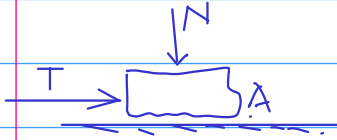


07/09/2020

Failure of Rocks

Friction on Rock Surfaces:-

Amontons's law:- (1699)



$$T = \mu N.$$

μ = coeff. friction.

Shearing / traction

$$\frac{T}{A} = \mu \frac{N}{A} \Rightarrow \boxed{\tau = \mu \sigma}$$

μ = coeff. Static friction.

Cond'n. for
maintaining motion

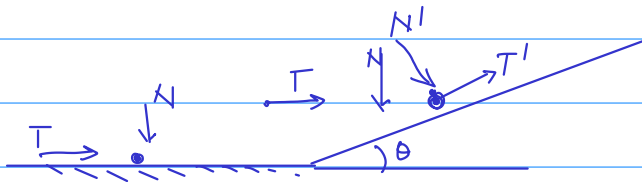
$$\tau = \mu_d \sigma$$

$$\mu_d < \mu$$

μ_d = dynamic coeff. of friction.

Coulomb (1785)

For irregular surfaces it may happen that sliding takes place up an inclined plane by some angle θ to the Nominal plane of Contact.



$$N' = N \cos \theta + T \sin \theta$$

$$T' = T \cos \theta - N \sin \theta.$$

$$T' = \mu N'$$

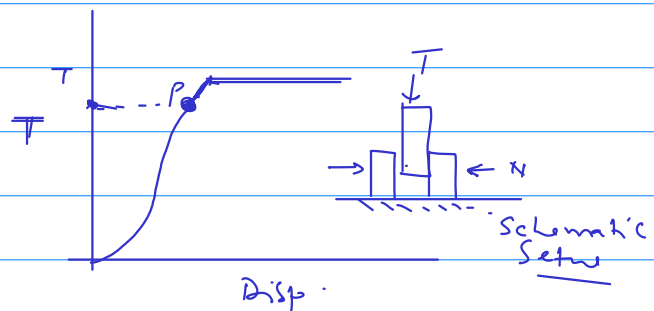
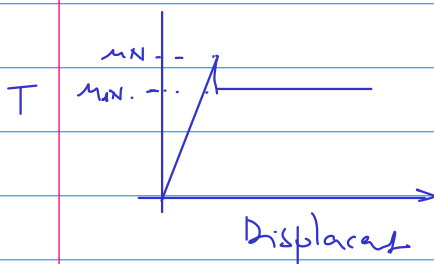
$$T = \frac{N(\mu + \tan \theta)}{(1 - \mu \tan \theta)}$$

$$\theta \ll 1$$

$$T = \frac{N(\mu + \theta)}{(1 - \mu \theta)} = N(\mu + \theta)(1 + \mu \theta)$$

$$T = N[\mu + (1 + \mu^2)\theta] \quad \theta^2 \text{ term} \rightarrow 0$$

Hoskin-1968

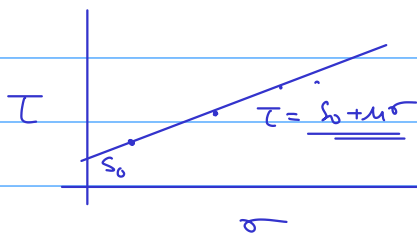


Trachyte.

Creation of new sliding surface $\rightarrow$ slickensides and detrital material.
(Breakage of asperities)



$$\tau = \mu \sigma$$

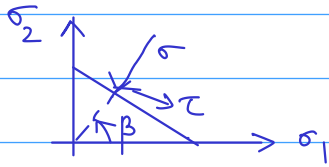


$$\tau = S_0 + \mu \sigma$$

Coulomb.
1785

$$\begin{aligned} \mu &\rightarrow T = \mu N \\ \mu &\rightarrow \tau = S_0 + \mu \sigma \end{aligned}$$

Sliding on a plane of weakness



Criteria for Slipping is

$$|\tau| = S_0 + \mu \sigma$$

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_2) + \frac{1}{2}(\sigma_1 - \sigma_2) \cos 2\beta$$

$$\tau = -\frac{1}{2}(\sigma_1 - \sigma_2) \sin 2\beta$$

$$\sigma_m = \frac{1}{2}(\sigma_1 + \sigma_2) = \text{Mean Normal stress}$$

$$\tau_m = \text{max shear stress} = \frac{1}{2}(\sigma_1 - \sigma_2)$$

$$\mu = \tan \phi$$

$$\tau_m \sin 2\beta = S_0 + \mu [\sigma_m + \tau_m \cos 2\beta]$$

$$\tau_m = \frac{(\sigma_m \tan \phi + S_0)}{\sin 2\beta - \tan \phi \cos 2\beta}$$

$$= \frac{(\sigma_m \tan \phi + S_0) \cos \phi}{\sin 2\beta \cos \phi - \sin \phi \cos 2\beta}$$

$$= \frac{\sigma_m \sin \phi + S_0 \cos \phi}{\sin(2\beta - \phi)}$$

$$\boxed{\tau_m = \frac{(\sigma_m + S_0 \cot \phi) \sin \phi}{\sin(2\beta - \phi)}}$$

$$\frac{\sigma_1 - \sigma_2}{2} = \frac{\left[\frac{\sigma_1 + \sigma_2}{2} + S_0 \cot \phi \right] \sin \phi}{\sin(2\beta - \phi)}$$

$$\begin{aligned}
 (\sigma_1 - \sigma_2) \sin(2\beta - \phi) &= (\sigma_1 + \sigma_2) \sin \phi + 2 S_0 \cot \phi \sin \phi \\
 &\quad + 2\sigma_2 \sin \phi - 2\sigma_2 \sin \phi \\
 &= (\sigma_1 - \sigma_2) \sin \phi + 2\sigma_2 \sin \phi + 2 S_0 \cos \phi
 \end{aligned}$$

$$(\sigma_1 - \sigma_2) [\sin(2\beta - \phi) - \sin \phi] = 2(\sigma_2 \sin \phi + S_0 \cos \phi)$$

$$\sigma_1 - \sigma_2 = \frac{2(\sigma_2 \sin \phi + S_0 \cos \phi)}{\sin(2\beta - \phi) - \sin \phi}$$

$$= \frac{2(\sigma_2 \sin \phi + S_0 \cos \phi)}{2 \cos \beta \sin(\beta - \phi)}$$

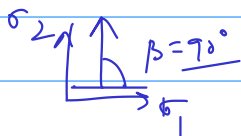
$$= \frac{2(\sigma_2 \sin \phi + S_0 \cos \phi) / \cos \phi}{2 \cos \beta [\sin \beta \cos \phi - \cos \beta \sin \phi] / \cos \phi}$$

$$= \frac{2(S_0 + \mu \sigma_2)}{2 \cos \beta [\sin \beta - \mu \cos \beta]}$$

$$\sigma_1 - \sigma_2 = \frac{2(S_0 + \mu \sigma_2)}{(1 - \mu \cot \beta) \sin 2\beta}$$

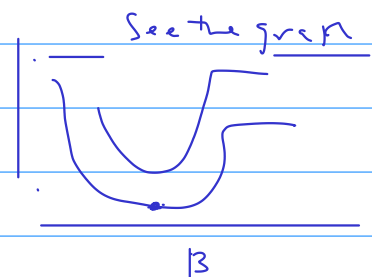
①

$$\frac{d}{d\beta} \left(\frac{\mu}{\nu} \right) = \frac{\nu \frac{d\mu}{d\beta} - \mu \frac{d\nu}{d\beta}}{\nu^2}$$



$$\sigma_1 - \sigma_2 \rightarrow \infty$$

(\sigma_1 - \sigma_2)



$$\beta = \phi \quad \sigma_1 - \sigma_2 \rightarrow \infty$$

$$\phi < \beta < \frac{\pi}{2}$$

$$\tan 2\beta = -\frac{1}{\mu}$$

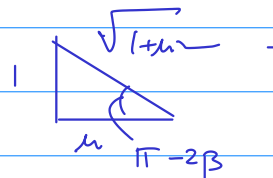
$$\sigma_1 = \sigma_2 + \frac{2(\sigma_0 + \mu\sigma_2)}{(1 - \mu \cot \beta) \sin 2\beta}.$$

$$= \sigma_2 + \frac{2(\sigma_0 + \mu\sigma_2)}{\sin 2\beta - \mu \times 2 \cos^2 \beta}.$$

$$= \sigma_2 + \frac{2(\sigma_0 + \mu\sigma_2)}{\sin 2\beta - \mu(1 + \cos 2\beta)}.$$

$$-\tan 2\beta = +\frac{1}{\mu}.$$

$$\tan(\pi - 2\beta) = \frac{1}{\mu}.$$



$$\sin 2\beta = \sin(\pi - 2\beta) = \frac{1}{\sqrt{1+\mu^2}}$$

$$\cos(\pi - 2\beta) = \frac{\mu}{\sqrt{1+\mu^2}}$$

$$\cos 2\beta = \left(\frac{-\mu}{1+\mu^2} \right)$$

$$\sigma_r = \sin 2\theta - \mu(1 + \cos 2\theta)$$

$$= \frac{1}{\sqrt{1+\mu^2}} - \mu \left(1 - \frac{\mu}{\sqrt{1+\mu^2}} \right).$$

$$= \frac{1 - \mu(\sqrt{1+\mu^2} - \mu)}{\sqrt{1+\mu^2}}.$$

$$= \frac{1 + \mu^2 - \mu\sqrt{1+\mu^2}}{\sqrt{1+\mu^2}} = \frac{\sqrt{1+\mu^2}[\sqrt{1+\mu^2} - \mu]}{\sqrt{1+\mu^2}}$$

$$= [\sqrt{1+\mu^2} - \mu]$$

$$\sigma_1 = \sigma_2 + \frac{2(S_0 + \mu \sigma_2)}{\sqrt{1+\mu^2} - \mu}$$

$$m. \quad \boxed{\sigma_1 = \sigma_2 + 2(S_0 + \mu \sigma_2) [\mu + \sqrt{1+\mu^2}]}$$

$$\beta = \frac{1}{2} \tan^{-1} \left(-\frac{1}{\mu} \right)$$

Attendance-31

09/09/2020 Coulomb failure Criteria

Friction analysis - $|\tau| = S_0 + \mu \sigma$

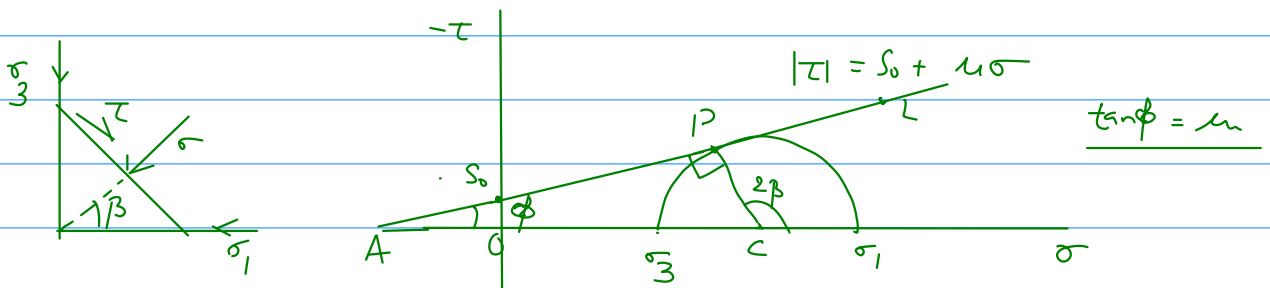
$\uparrow$
cohesion

$\sqrt{1+\mu^2}$

Intact Rock $|\tau| = S_0 + \mu \sigma$

$\boxed{\tau} \leftarrow \sigma$

(τ, σ) —



$$2\beta = 180 - \angle APC$$

$$\angle APC = \pi - 2\beta$$

$$\angle CPA = \frac{\pi}{2}$$

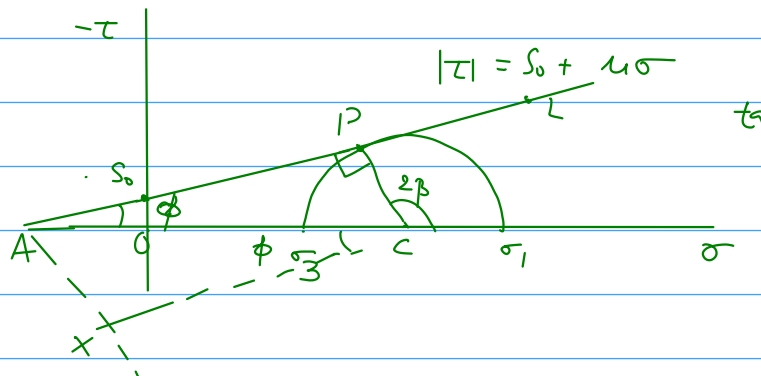
$$\angle PAC = \phi$$

$$\pi = -2\beta + \pi + \frac{\pi}{2} + \phi$$

$$2\beta = \frac{\pi}{2} + \phi$$

$$\boxed{\beta = \frac{\pi}{4} + \frac{\phi}{2}}$$

$$\boxed{\beta = \frac{\pi}{4} - \frac{\phi}{2}}$$



$$\sin \phi = \frac{Ax}{Ac} = \frac{PC}{Ac}$$

$$CP = Ac \sin \phi$$

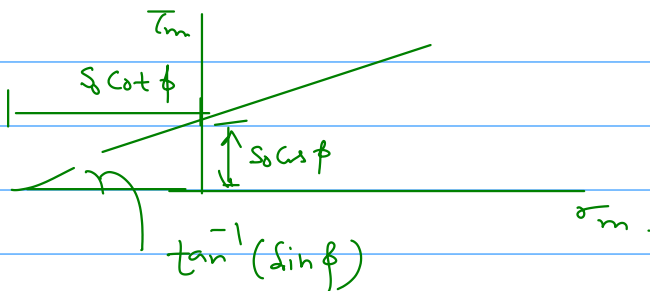
$$= (AO + OC) \sin \phi = AO \sin \phi + OC \sin \phi$$

$$= s_0 \cot \phi \cdot \sin \phi + OC \sin \phi = s_0 \cos \phi + OC \sin \phi$$

$$\frac{1}{2} (\sigma_1 - \sigma_3) = s_0 \cos \phi + \frac{1}{2} (\sigma_1 + \sigma_3) \sin \phi$$

$$\sigma_m = \frac{1}{2} (\sigma_1 + \sigma_2) \quad \tau_m = \frac{1}{2} (\sigma_1 - \sigma_3)$$

$$\tau_m = s_0 \cos \phi + \sigma_m \sin \phi$$



Rearranging the $(\sigma_1 - \sigma_3)$ Eqn

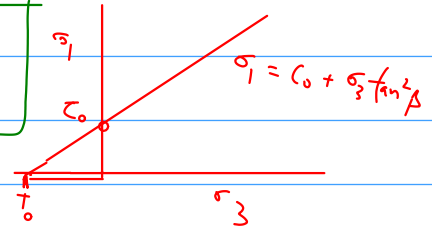
$$\frac{1}{2} (\sigma_1 - \sigma_3) = s_0 \cos \phi + \frac{1}{2} (\sigma_1 + \sigma_3) \sin \phi$$

$$(\sigma_1 - \sigma_3) = 2s_0 \cos \phi + (\sigma_1 + \sigma_3) \sin \phi$$

$$\sigma_1(1 - \sin \phi) = 2S_0 \cos \phi + \sigma_3(1 + \sin \phi)$$

or,

$$\sigma_1 = \frac{2S_0 \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$$



$$\beta = \frac{\pi}{4} + \frac{\phi}{2}$$

$$\begin{aligned} 1^{\text{st}} \text{ Coeff} &= \frac{\cos \phi}{1 - \sin \phi} = \frac{\cos(2\beta - \frac{\pi}{2})}{1 - \sin(2\beta - \frac{\pi}{2})} = \frac{\sin 2\beta}{1 + \cos 2\beta} = \frac{2 \sin \beta \cos \beta}{2 \cos^2 \beta} \\ &= \tan \beta \end{aligned}$$

2nd Coeff

$$\frac{1 + \sin \phi}{1 - \sin \phi} = \tan^2 \beta$$

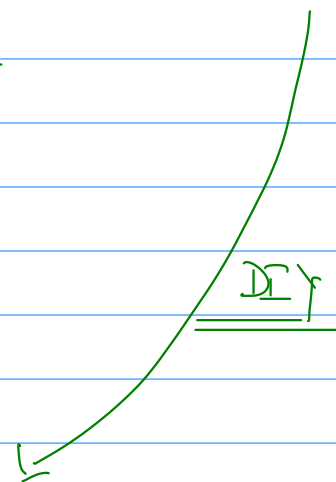
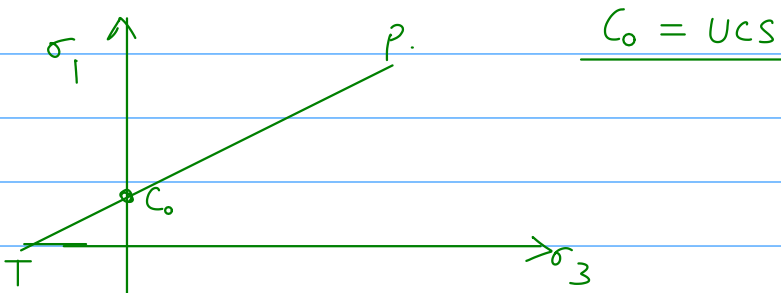
$$\sigma_1 = 2S_0 \tan \beta + \sigma_3 \tan^2 \beta$$

or,

$$\sigma_1 = C_0 + \sigma_3 \tan^2 \beta$$

$$\mu = \tan \phi$$

$$\cos \phi = \frac{1}{\sqrt{1+\mu^2}}, \sin \phi = \frac{\mu}{1+\sqrt{1+\mu^2}}$$



$$\sqrt{1+\mu^2} \cdot \tau_m = S_0 + \mu \sigma_m$$

$$\sigma_1 = 2S_0 [\mu + \sqrt{1+\mu^2}] + [\sqrt{1+\mu^2} + \mu]^2 \sigma_3$$

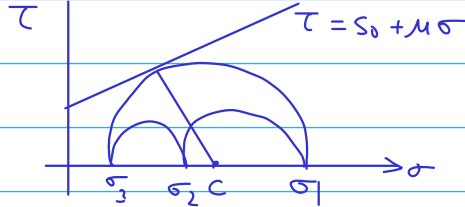
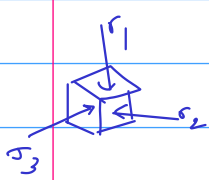
$$UCS = 2S_0 [\mu + \sqrt{1+\mu^2}]$$

Attendance - 18

11/09/2020

Discussion on Coulomb's Theory

In 3D π Mohr's Circle is



The first Mohr's Circle to touch the failure line will be that corresponding to σ_1 & σ_3 .

$$\sigma = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}$$

$$\sigma_x + \sigma_y + \sigma_z = 1$$

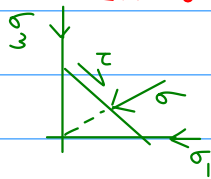
$\Rightarrow$ From Coulomb's assumption $\rightarrow \sigma_2$ has no role to play in failure of rock.

For many years Experiments were conducted in $\sigma_2 = \sigma_3$ environment.

Recent experiments conducted in true-triaxial conditions $\rightarrow$ for some rocks the assumption of Coulomb is not valid.

(2)

In Coulomb's Hypothesis it is considered that normal stress over the failure plane is +ve (ie. compressive).



$\Downarrow$
We need to ignore certain portion of the Coulomb's (σ_1 vs σ_3 Curve).

Let's see

Condition of positivity of σ is

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_3) + \frac{1}{2}(\sigma_1 - \sigma_3) \cos 2\beta > 0$$

$$\text{using } \beta = \frac{\pi}{4} + \frac{\phi}{2}$$

$$\frac{1}{2}(\sigma_1 + \sigma_3) + \frac{1}{2}(\sigma_1 - \sigma_3) \sin \phi > 0$$

Re-arranging the sides

$$\sigma_1(1 - \sin \phi) > -\sigma_3(1 + \sin \phi)$$

$$\text{or } \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right) > -\sigma_1$$

Compare with

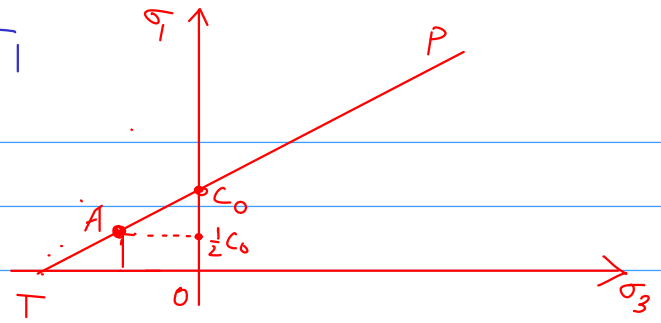
$$\sigma_1 = \frac{2s_0 \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$$

$$\& \quad \sigma_1 = 2s_0 \tan \beta + \sigma_3 \tan^2 \beta$$

$$\Rightarrow \sigma_1 > \frac{2S_0 \cos \phi}{1 - \sin \phi} - \sigma_1$$

$$\text{or } \sigma_1 > s_0 \tan \beta.$$

$$\text{or } \boxed{\sigma_1 > \frac{1}{2} C_0}$$



The portion TA of the failure line TAP does not correspond to the +ve normal stress on failure plane. (Rather it is tensile)

$\therefore$ The portion TA does not properly predict the shear failure.

Concept of Tensile Cut-off (After Paul 1961)

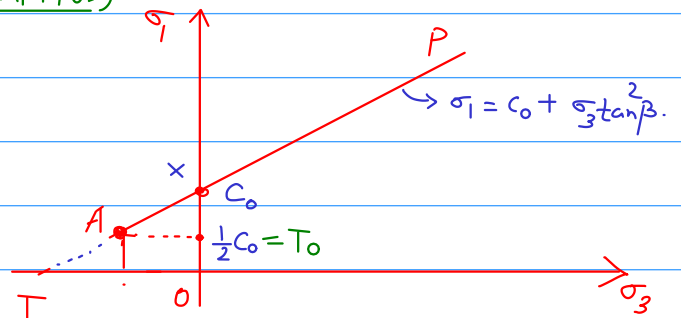
The simplest modification could

be Truncate TAP at point A

corresponding to $\sigma_1 = \frac{C_0}{2}$ & extend

it as a vertical line till it

meets σ_3 line at T_0



Failure in the region TA will lead to extensional failure

$\therefore$ Hence, the $UTS = T_0 = \frac{1}{2} C_0$. ✓ (Theoretically $\frac{C_0}{T_0} = 2$ ✓ Coulomb's Assumption)

However, experimental data shows $\frac{C_0}{T_0} > 2$.

$\therefore$ Paul (1961) suggested a bilinear curve for Coulomb Criteria using experimentally measured value of T_0 .

$$\begin{aligned} \sigma_1 &= C_0 + \sigma_3 \tan^2 \beta & \forall \sigma_1 > C_0 \left[1 - \frac{C_0 T_0}{4 S_0^2} \right] \\ \sigma_1 &= T_0 & \forall \sigma_1 < C_0 \left[1 - \frac{C_0 T_0}{4 S_0^2} \right] \end{aligned}$$

Background for Mohr's Hypothesis

The limitations in Coulomb's Hypothesis to explain the experimental results led Mohr to Develop a different Hypothesis.

The Coulomb's Hypothesis States that

- Failure will occur on a plane when the normal stress and Shear Stress satisfy the condition

$$\tau = s_0 + \mu \sigma$$



Hence, the condition appears to be a straight line in $(\sigma, |\tau|)$ -plane with slope μ .

The straight line could not be seen in many experiments.

- The theory ignores the effect of the intermediate principal stress - recent researches in true triaxial conditions reveal indeed σ_2 plays a role.

- Within the CONTEXT of Coulomb's Theory, The UTS can be found by setting $\sigma_1 = 0$ in the Eqn. ↓ (Tensile strength)

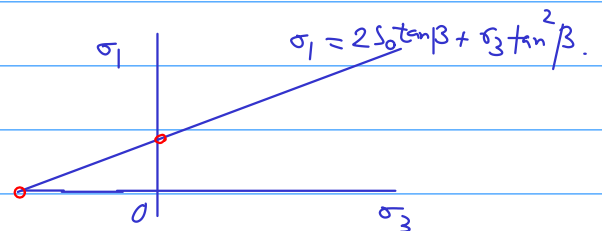
$$\sigma_1 = 2s_0 \tan \beta + \sigma_3 \tan^2 \beta$$

$$\text{or, } 0 = 2s_0 \tan \beta + \sigma_3 \tan^2 \beta$$

$$\text{or, } -\sigma_3 = 2s_0 \cot \beta$$

$$\text{or, } T_0 = 2s_0 \cot \beta$$

$$\text{or, } T_0 = \frac{2s_0}{(\mu + \sqrt{1+\mu^2})}$$



& UCS:- can be found by setting $\sigma_3 = 0$

$$\Rightarrow \sigma_1 = 2s_0 \tan \beta$$

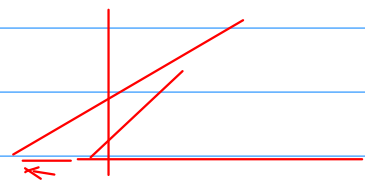
$$\text{or, } c_0 = 2s_0 \tan \beta$$

Hence the Coulomb's Theory predicts that

$$\frac{c_0}{T_0} = \tan^2 \beta = [\mu + \sqrt{1+\mu^2}]^2$$

$\forall \mu = \text{very small value}$

$$\frac{c_0}{T_0} < 1$$



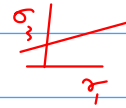
$$\forall \mu = 1 \quad \frac{c_0}{T_0} = 5.83$$

Experimental values of this ratio tends to be of the order 10 or so.

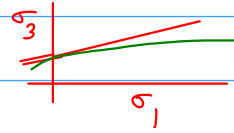
(Roughly this deficiency can be expressed by saying that the Coulomb failure line extends too far into the tensile region of (τ, σ) plane. or (σ_1, σ_3) plane).

- ④ Coulomb's Theory also predicts that the Compressive Stress required to cause failure, i.e. σ_1 increases linearly with Confining Stress.

Corollary of point-①.



Experiments typically show that σ_1 at failure will increase at a less than linear rate with σ_3



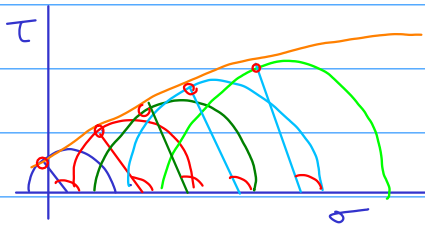
Mohr's Proposition to correct aforementioned deficiencies (1900)

Replace the linear relation $|T| = \underline{s_0 + \mu \sigma}$



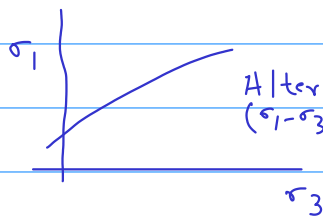
Non Linear Generalised Curve $|T| = f(\sigma)$.

The non-linear curve can be determined experimentally.



Features

- ① The basic idea of σ_2 (ineffective) retained.
- ② Concave downwards so $\beta \downarrow$ with $\uparrow \sigma_3$.
- ③ Orientation of failure plane can be predicted by Mohr Circle construction.



Alternatively $(\sigma_1 - \sigma_3)$ plot can also be made.

Empirical failure Criteria.

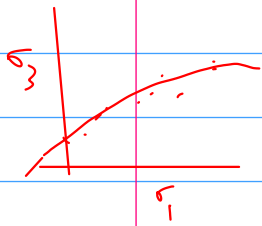
Based on Mohr's Proposition $\rightarrow$ Various mathematical formula has been proposed.
to fit the experimental data.

so, all empirical formulae have (based on Mohr's Hypothesis)
have forms as under:

$$\left. \begin{aligned} |T| &= f(\sigma) \\ \sigma_1 &= g(\sigma_3) \end{aligned} \right\}$$

Hoek & Brown failure law:
(Hoek 1980).

Revised - 2018 (GSI)



$$\sigma_1 = \sigma_3 + (m \sigma_c \sigma_3 + \sigma_c^2)^{1/2}$$

m & σ_c are two fitting parameters

Set $\sigma_3 = 0 \Rightarrow \sigma_1 = \sigma_c = \text{UCS} \checkmark$

Set $\sigma_1 = 0 \Rightarrow T_0 = -\sigma_3 = \frac{\sigma_c}{2} [\sqrt{m^2 + 4} - m]$

The parameter $5 < m < 30$

Large values of m

T_0 approximates to $T_0 = \frac{\sigma_c}{m}$

$\Rightarrow \frac{\sigma_c}{T_0} = \frac{C_0}{T_0} = m$

$\therefore$ HB-Model Predicts larger $\frac{C_0}{T_0}$ values than Coulomb model.

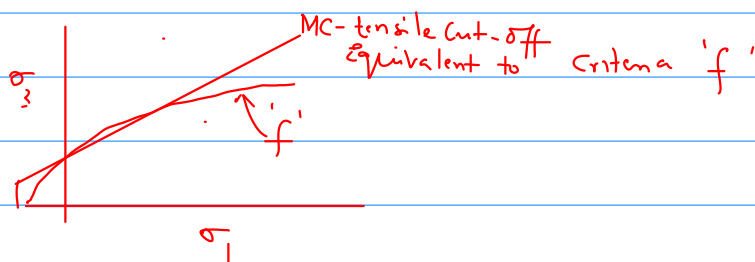
$m \approx 7$ Carbonate rocks with well developed cleavage.

$m \approx 10$ lithified argillaceous rock (mudstone, shale etc).

$m \approx 17$ fine grained polyminerallic crystalline
Igneous rocks

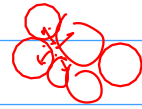
$m \approx 25$ Coarse-grained igneous rocks, (gabbro, gneiss,
Granite).

Why is linear fit normally given with empirical formula?



Effects of Pore Pressure

Effects of pore fluid
— mechanical
— Chemical.



* Pore-pressure acts "outward" from the pore space.
It is same in all directions.

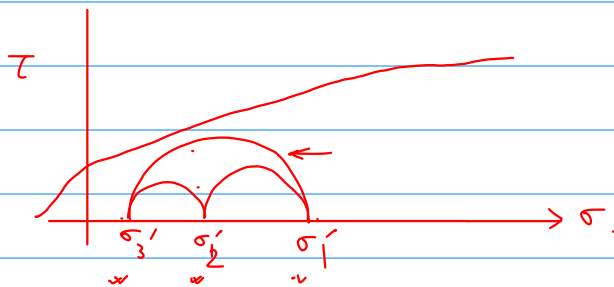
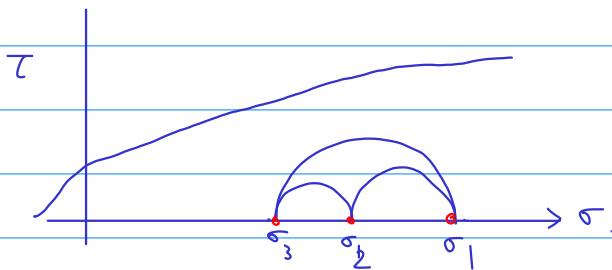
* Using this reasoning Karl Terzaghi (1936)

↓ Proposed

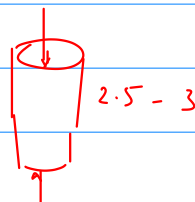
Failure of Soil will be controlled by "Effective stress"

$$\text{ie } \sigma'_i = \sigma_i - p$$

$$\text{or, } \sigma'_1 = \sigma_1 - p, \sigma'_2 = \sigma_2 - p, \sigma'_3 = \sigma_3 - p.$$

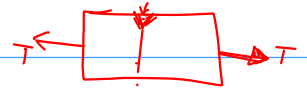


Explanation of Hill Slides/
land slides after Rain fall.



Griffith's Theory of failure

- micro mechanical model of failure
- Theory of fracture mechanics ✓
- 1924 → model for rocks with large no of flaws (elliptical)



Hypothesis:- Crack will extend in an unstable manner when max tensile stress at any point along the crack boundary reaches some value characteristic of rock material.

Result:

$$(\sigma_1 - \sigma_2)^2 = 8T_0(\sigma_1 + \sigma_2) \quad \forall \quad \sigma_1 + 3\sigma_2 > 0$$

$$\sigma_2 = -T_0 \quad \forall \quad \sigma_1 + 3\sigma_2 < 0$$

